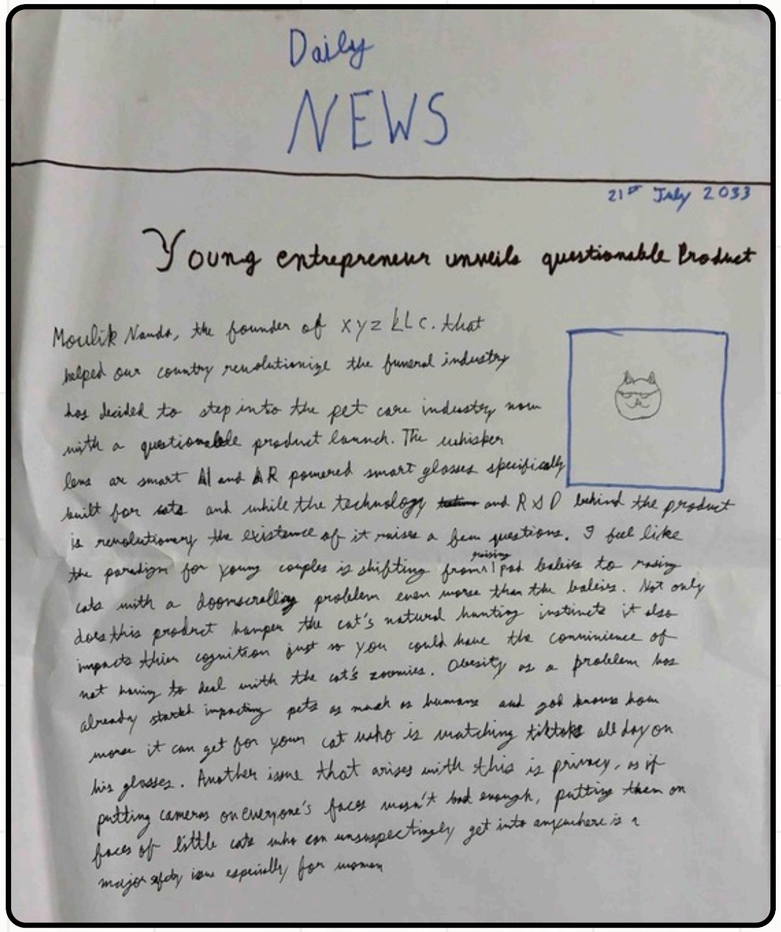
MOULIK NANDA

A3 PRESENTATION

THE MAKINGS OF A PERVERT DETECTOR

A Prelude to the Problem



Daily

NEWS

21st July 2033

Young entrepreneur unveils questionable product

Monlik Nando, the founder of xyz LLC. that helped our country revolutionize the funeral industry has decided to step into the pet care industry now with a questionable product launch. The whisker lens are smart AI and AR powered smart glasses specifically built for cats and while the technology ~~behind~~ and R&D behind the product is revolutionary the existence of it raise a few questions. I feel like the paradigm for young couples is shifting from ~~I got babies to raising~~ raising cats with a doomscrolling problem even worse than the babies. Not only does this product hamper the cat's natural hunting instincts it also impacts their cognition just so you could have the convinience of not having to deal with the cat's zoomies. Obesity as a problem has already started impacting pets as much as humans and god knows how worse it can get for your cat who is watching tiktoks all day on his glasses. Another issue that arises with this is privacy, as if putting cameras on everyone's faces wasn't bad enough, putting them on faces of little cats who can unsuspectingly get into anywhere is a major safety issue especially for women

news aticle written for the excercise,
excuse the handwritng

Right before the start of the studio during our orientation week, Vineeta made us participate in an exercise where we had to anticipate where we see ourselves 5 years from now and write a newspaper article in this hypothetical future talking about ourselves. Reading about all of the recent developments in tech, I was disillusioned and pessimistic about what a designer would be working on 5 years down the line that would end up in the news cycle.

Which is why, instead of writing a traditional article about me being a renowned designer 5 years in the future, I wrote a hit piece on future me for giving up on my values and morals by making “AI and AR powered smart glasses for cats” and speculated on all of the demerits of such a product.

The Beginning

Starting this studio by researching the Luddite movement and its foundations made me think about the recent developments in tech and my speculative article once again.
Further research on the topic helped me draw more parallels to the luddite movement while also realising that this problem is much bigger than what I initially thought. From being used by perverts to non-consensually record women to the glasses sending all recordings back to Meta, who would employ people in impoverished countries just to look at the recordings and train AI, there were way too many demerits to this product for it to exist in the first place, let alone be the smartphone replacement that Zuckerberg desperately wants it to be.

While the public perception of the glasses has gotten overly negative over the last few years, we still did not have a hammer for this server room and my next goal was to come up with it.

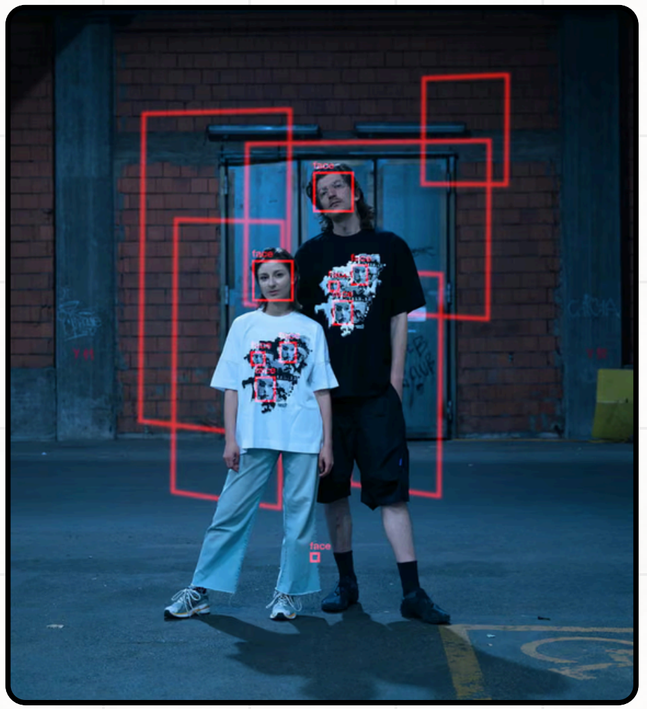


Always watching

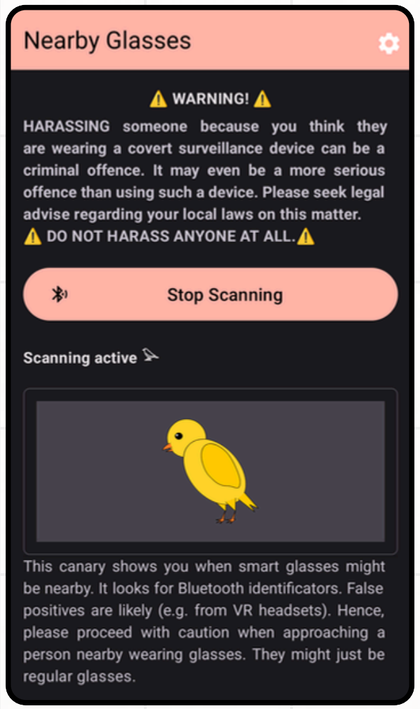
Ideation

The next goal for me was to continue my research and come up with ideas for the hammer that would help me “Defeat” the pervert glasses. After looking into pre-existing and more speculative solutions I came up with 3 ideas for my hammer.

1. A malicious QR code displayed in public spaces that the glasses scan automatically, this QR code could link to an audio/video message for the wearer or have something more malicious, like malware.
2. Clothing that uses carefully designed arrangements of shapes, colours and repeated motifs to exploit weaknesses in some computer vision systems to prevent facial scanning.
3. A physical device, that would analyse nearby Bluetooth signals and match them with known signals and inform people about the presence of the glasses around them.



Clothing designed to fool facial recognition models



An app that used Bluetooth signals to detect smart glasses

In the end, the third idea seemed the most ideal since the QR codes have been patched when this exploit was used on the Google Glass and anti-AI clothing did not solve the problem of glassholes recording you without consent.

Critical Making and Protocol

Before we went further along with our ideas and concepts, we were introduced to some new frameworks, being critical making and protocol. At the time, I was not able to properly grasp Matt Ratto's paper on critical making but in hindsight, my project did end up taking a lot of cues from the concept of critical making, especially with all of the discussions it spawned with the people around me and how it further informed the direction my project took.

My reading of protocol by Alexander R. Galloway and the chapter on tactical media introduced me to a brand new form of resistance and activism different from the luddite movement of the past, one where you exploit flaws in the protocol itself to your advantage. Examples would include computer viruses and the cyberfeminist movement. Once again in hindsight, tactical media ended up being more relevant to my project than I initially imagined. I believe that if I take my project further, open-source it and share how I made it online, it would take the form of tactical media against big corporations like Meta itself.

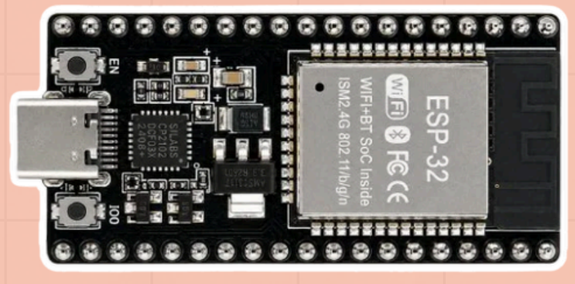
The Concept- CAP

CAP (short for Catch a Pervert) is my response to these glasses, my goal was to combine fashion and technology to create something that works against the precedent set by these glasses by showing the public that they have the power to push back and fight.

CAP was meant to be a device that detects the presence of any pair of smart glasses around you by scanning around for Bluetooth signals, looking for manufacturer and device IDs that match the glasses and let you know if there is a glasshole lurking nearby, putting control of your privacy back into your hands.

How do we make it?

While I had still not decided what form I wanted cappy to take I did know that no matter the form the core of cappy would remain the same and hence I started working on a parts list.



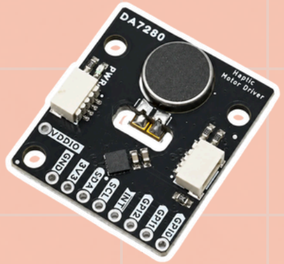
ESP 32



E-ink



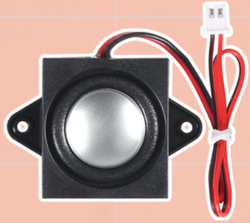
LCD



Haptic Motor



Dot Matrix



Speaker



Battery



Charger

first list of parts

My initial parts list was a bit ambiguos as there were still things on the list I had to pick between like a dot matrix vs e-ink vs lcd display and betwen a haptic motor or a speaker. Aditionally in further iterations to align with my form I switched to a smaller ESP 32 S3 zero and a USB C charging solution based on feedback.

Ideating on the Form

Since I wanted the product to be as fashionable as it is functional, picking a form was one of the most important decisions of this project. I started off with a mood board with a collection of products I was inspired by.

A list of my intial ideas included-

- an E-ink name tag
- A wearable accessory like a pendant or hair clip
- a keychain/clothes pin
- A watch/arm cuff

alongside these ideas in my moodboard i also came up with 2 more unconventional ideas-

- a hat with a police siren on it
- a mannequin at the entrances of public spaces with “laser eyes”



Finalisng the Concept

Parts

I came up with my final parts list based on my decisions, iterations and external feedback. Budget was also a consideration here since originally I was going with the E-ink name tag form but good quality E-ink displays being out of my budget pushed me into going for a round LCD display.

1	ESP32-S3 zero	BLE scanning + logic
2	Round TFT display module	1.28", 240x240
3	USB-C LiPo charger module	TP4056-based
4	LiPo battery	3.7V, 500-1000mAh
5	Push button/switch	on/off button
6	Breadboard	For testing purposes
7	Jumper wire	M-F, F-F, M-M
8	resistors kit	220ohm-10kohm
9	USB-C cable	Charging/testing

Form

After abandoning my first Idea for an e-ink name tag due to budget constraints and deciding that a round display would work better for my product, I went to Figma and designed the first 2d mockup of my Idea. I envisioned a modular wearable device taking inspiration from multiple items in my mood board, that uses magnets with different attachments that the user could connect the magnet to.



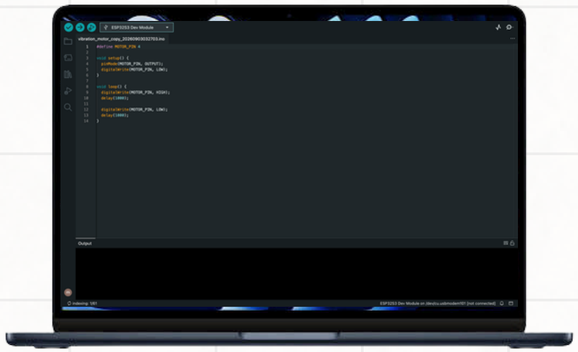
Form concept in multiple colour ways

Start of my Electronics Journey

I started out by doing a little shopping spree at SP Road, where I got my esp 32. After doing some independent research and asking around friends who have experience with electronics. I was able to set up my ESP 32 and run my first peice of code on it, which would light up the LED on the ESP and cycle the colours through the RGB spectrum. Setting this up opened a bunch of doors for me.

For the next step, I wanted to validate if this ESP would be able to detect and isolate Bluetooth signals around it so that the detection feature of my product could function. I started by explaining my product to an LLM and asked it to give me the Bluetooth detection code that would detect my apple watch but my first few attempts were a failure due to my lack of knowledge of electronics and modern Bluetooth protocols.

This project would not be as easy as I thought





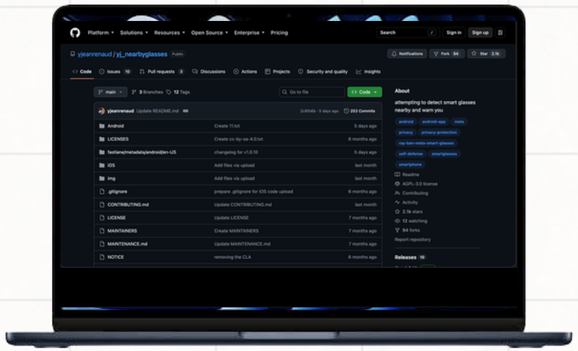
Trying Again

Picking up where I left off, I started again by doing more research and looking into the subject. Modern devices don't continuously advertise themselves in a simple, identifiable way. Especially products from renowned companies like apple and meta (even if meta doesn’t care about your own personal safety).

I needed a better method. It was time to look into **open source software.** Nearby Glasses, an Android app that functions as a detector for smart glasses around you has open sourced their code on GitHub, this project continuously sniffs nearby BLE advertisements to detect smart glasses and AR wearables using parameters like company identifiers and device name keywords and alerts you based on proximity. I adapted this code to my ESP so that the LED on it will switch colours from green to yellow to red based on the proximity of the glasses to the device.

Once the code was functional, I also modified it and added my earphones to the list of devices that the ESP is constantly scanning for so that I could test it in the real world.

[Link to testing detection model]





The Birth of Cappy

Once I had the detection code running on my ESP successfully, my next step was to turn it into an actual product, starting with the interface. I took inspiration from one of the devices on my moodboard- aptly named lilguy and decided to create a companion creature of my own that will live inside my product.

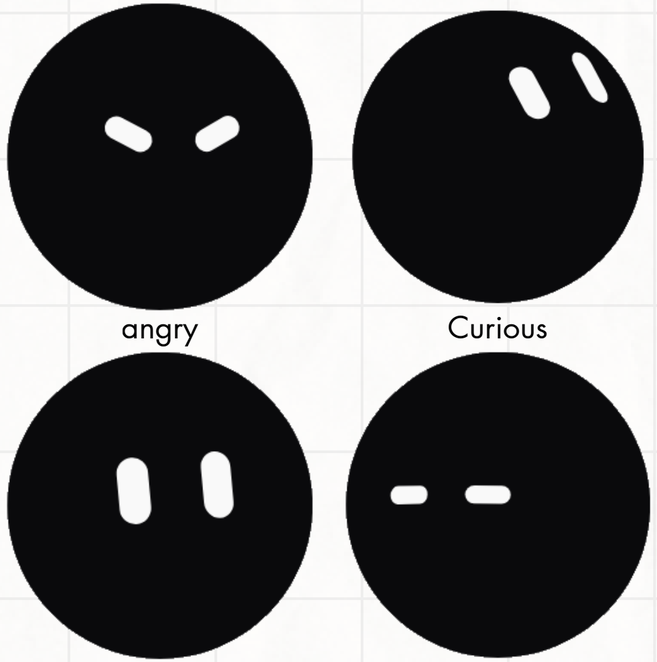
And this is how Cappy was born. I intended Cappy to be a simple, cute and expressive creature living inside the product that would react based on the proximity of the meta glasses around them, with the expressions switching from happy to curious to angry as someone wearing the glasses got closer. Alongside the change in expressions, the eyes of Cappy would also change colour from white to red to yellow as an additional visual indicator. As a fun extra, Cappy will start getting sleepy when it doesn't detect the glasses for an extended period of time.

After getting the visuals and animations of Cappy right, I exported them as gifs and with the help of google's antigravity AI, recreated the animation in code that could run on the ESP easily.

[Link to testing cappy and it's animations]



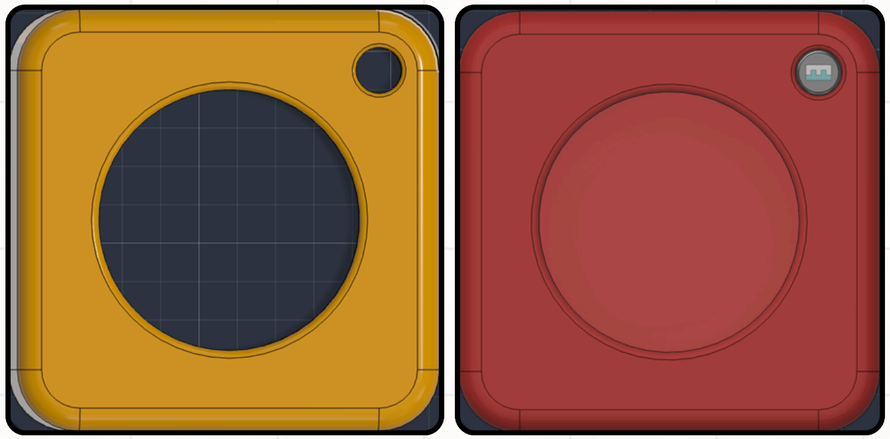
lilguy.net
the inspiration behind cappy



some of the initial visuals and animations I worked on for cappy

Giving Cappy a Body

I knew exactly how I wanted the body of Cappy to look based on my figma 2d renders. A squircle with a circular hole in the middle for the screen and another hole at the top right for the button. But making that simple design took way more time and effort than I initially thought.
I'm familiar with 3D modelling with Tinkercad, but for this project I wanted something more robust.
Which is how I ended up spending way more time than I planned on learning Fusion 360 with the help of YouTube. I measured all of my electronics with a measuring tape and used those measurements to make the design as compact as possible while still fitting in all of the parts and having some extra tolerance just in case. Unfortunately, I forgot to factor in the space that my jumper wires will take up in the case, making it extremely hard to fit in just the ESP connected to the display. On top of that I did not add a slot for the USB port on the case (I planned on getting it laser cut but wasn't able to), due to which I had to keep the case open to plug in the ESP 32 and use cappy.



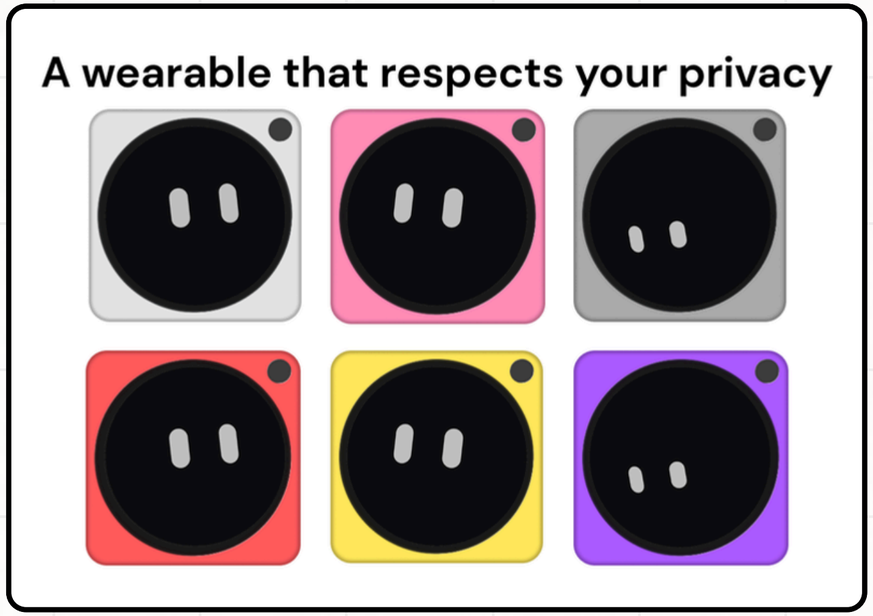
Cappy's body in the fusion 360 viewport



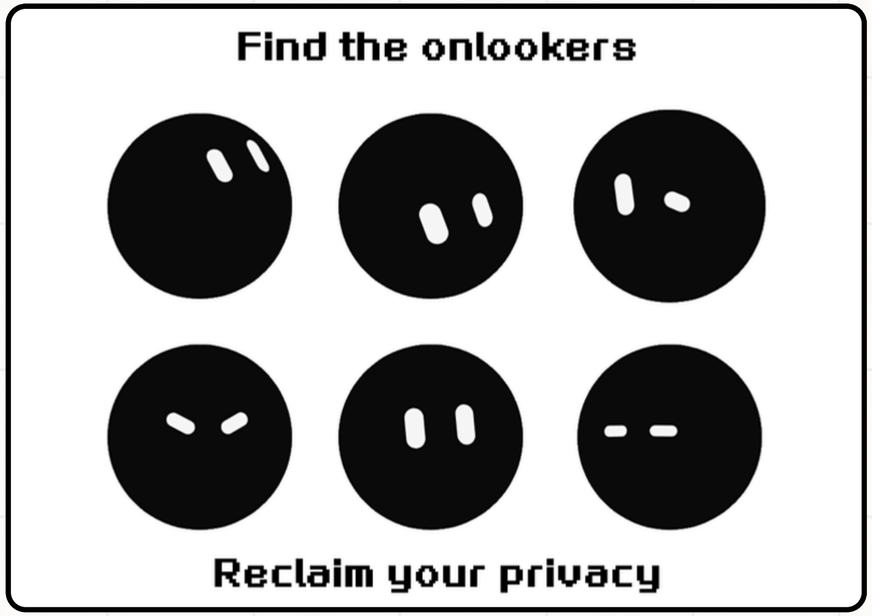
Final print of cappy V1

A2 Presentation and Posters

After finishing work on my prototype for CAP, it was time to get Cappy ready for my A2 presentation. This included working on a poster, the first iteration of which I didn't like and scrapped. I wanted Cappy to be the main focus of this presentation and steal the show with his personality, which is why I went back to the drawing board and designed a new poster primarily focusing on Cappy. Now Cappy was ready to be shown to the world in my presentation, where I demoed it using my Earphones. As of now, the prototype could work as both a design prototype and a functional prototype but could not work as both at the same time since the 3D printed enclosure could not be closed while the prototype was functional.



First poster designed for A2 presentation (scrapped idea)



Final poster used for A2 presentation

Post Prototype Plan

Based on the feedback on my prototype presentation and my own goals, these are the things I wanted to focus on going forward:-

1. Make Cappy more interactive.
2. Make the device battery powered.
3. Incorporate a vibration motor into CAP so that the user is aware of meta glasses wearers around them even when they aren't looking at Cappy.
4. Procure the actual glasses for the final submission presentation
5. Try to shrink the form factor of Cappy further down to make it smaller.

The first three points were relatively straightforward. The fourth would require more social outreach and a bit of luck, and the final one is the hardest. Shrinking down Cappy with my current hardware seemed impossible, I would need to replace all of my parts with more miniaturised and integrated parts (for example, an esp32 that has a display and accelerometer built in). That would mean that I have to start over and spend twice as much on the newer electronics. After further discussion with Manuel during my individual feedback session, I decided to go in the other direction and make it slightly bigger for both better functionality and visibility.

More Electronics!

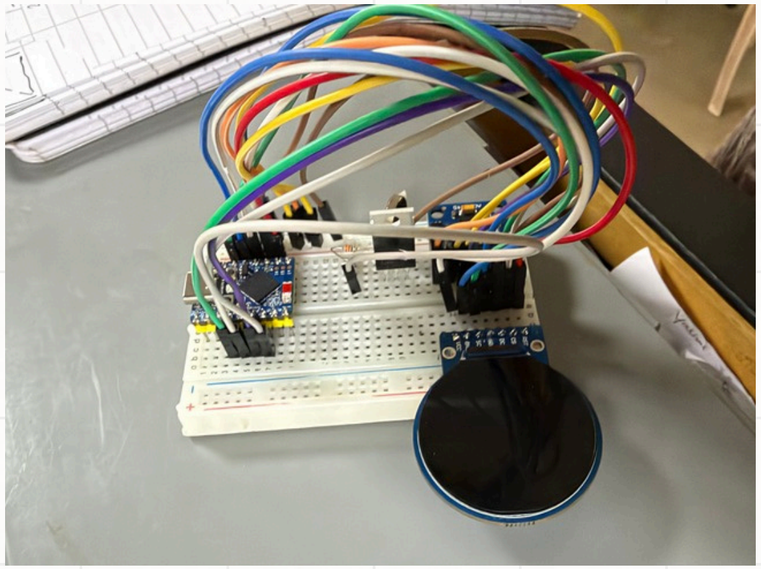
I went back to the breadboard and hooked up the following parts to it on top of my ESP and screen-

- An accelerometer with tap detection
- A vibration motor
- A mosfet to actually run the vibration motor.

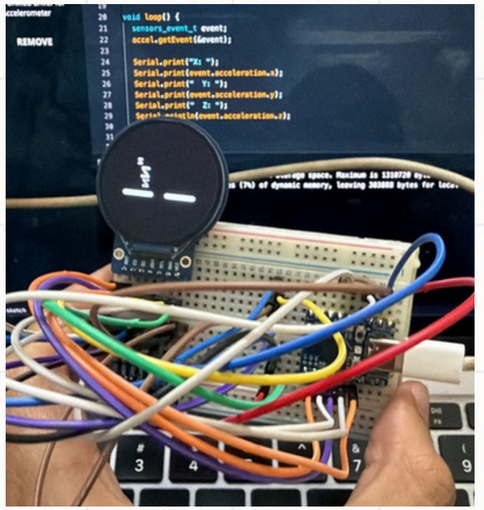
Once everything was wired up, I tested and calibrated the tap and double-tap detection on the accelerometer and tested the functionality of the vibration motor before I incorporated these into my project.

The vibration motor served as an additional output method other than the screen. It would do a sharp bzz when Cappy went into curious mode and a bzzbzz when switching to angry mode, so the wearer would always know what is up.

The accelerometer was the key to turning Cappy into an interactive companion. I integrated it as a “knock detector”, so when you would knock on the device while Cappy is sleepy or fully asleep he would wake up, albeit a little groggy.



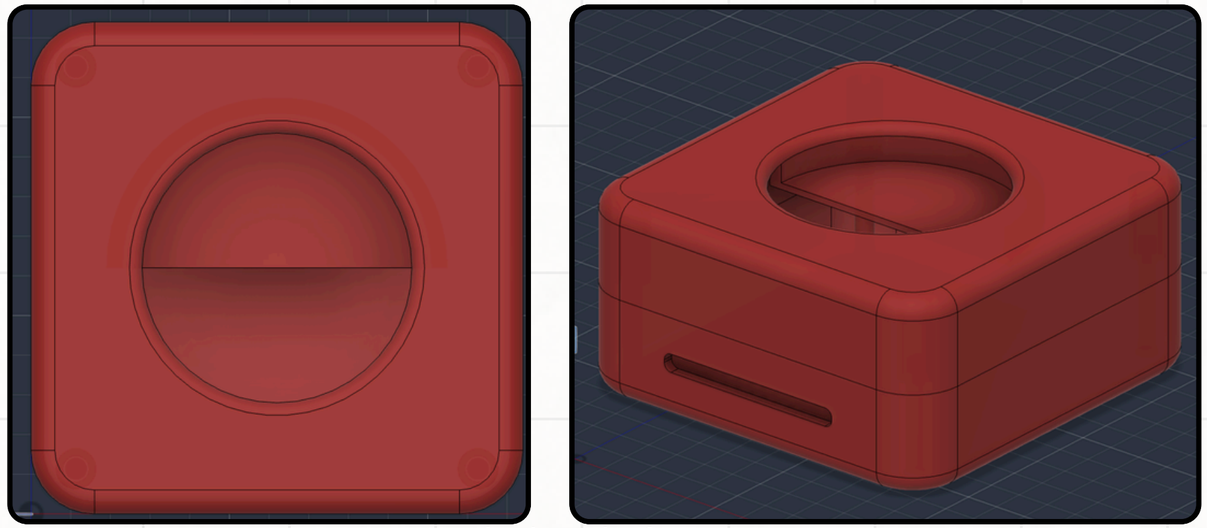
wiring all the electronics together



Testing the eletronics with Cappy

New Body, Same Old Cappy

With the new electronics, Cappy has outgrown its old body and as decided earlier, it was time to go big. Cappy 2.0 is bigger and better in every conceivable way (aside from portability). The way the top and bottom halves of the product connect has been redesigned so that this version stays shut, the screen was given much more support from the get go and instead of having space for a button, this version has a slot for an on/off switch and the USB-C charging port.



Cappy's new body in the fusion 360 viewport

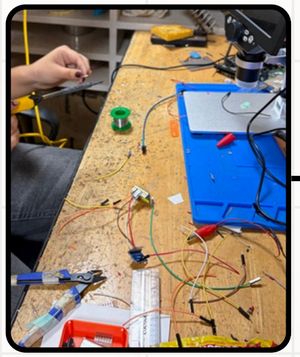


Cappy 2.0

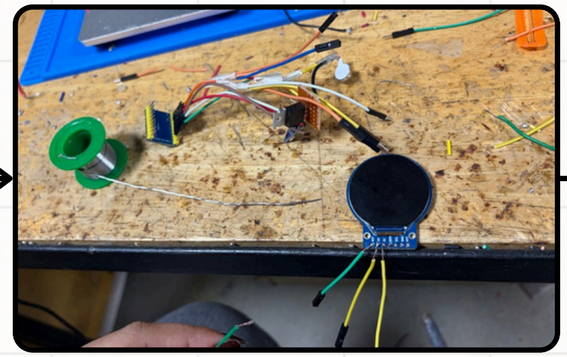
Putting it all Together

Now that Cappy's new body was ready, it was time to move him out of the breadboard and into the body. Which meant that it was finally time to do what I had been dreading- soldering. Without the support of Nataraju and peers like Sudarsini, I would have fried my ESP 32 in this process, so shoutout to them.

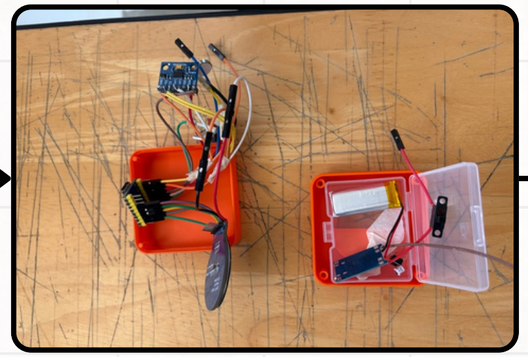
soldering timeline



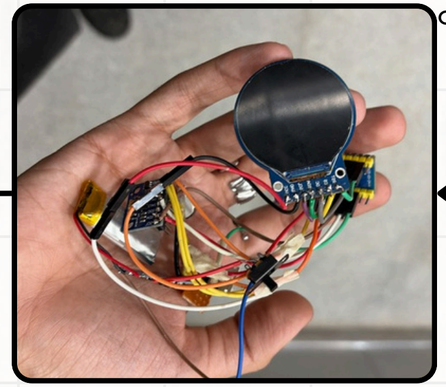
Soldering accelerometer and vibration motor



Soldering the display



Figuring out circuit for battery, charging system and switch next



All soldered



it fits!!

Mind and Body, Together at Last!









Giving Cappy the photoshoot he deserves

Struggle with Getting the Meta Glasses

Now that my product was nearly finished, there was only one thing left- procuring the Meta glasses for my A3 demonstration. This ended up being a bigger roadblock than I initially imagined. I started by reaching out to Jayashima, a professor on campus who might have the glasses. But I never heard back from him, even after sending a follow-up. Additionally, I also reached out on marketplace groups and subreddits on WhatsApp and Reddit, which did lead to 2 leads but one of them was concerned about lending something this expensive to a stranger on Reddit, and the other one lived in JP Nagar and disappeared the second I mentioned Yelahanka.

After failing to procure them via these methods, the next step was for me to utilise even shadier methods to procure never-before-seen, unreleased versions of the glasses that I would use for my A3 presentation demonstration.

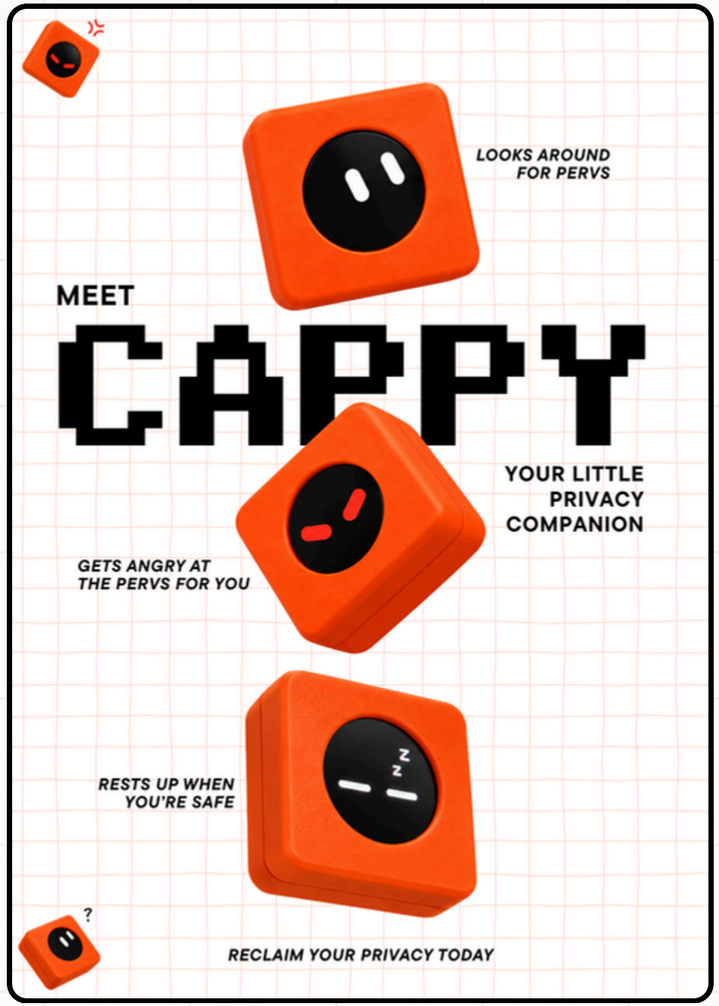


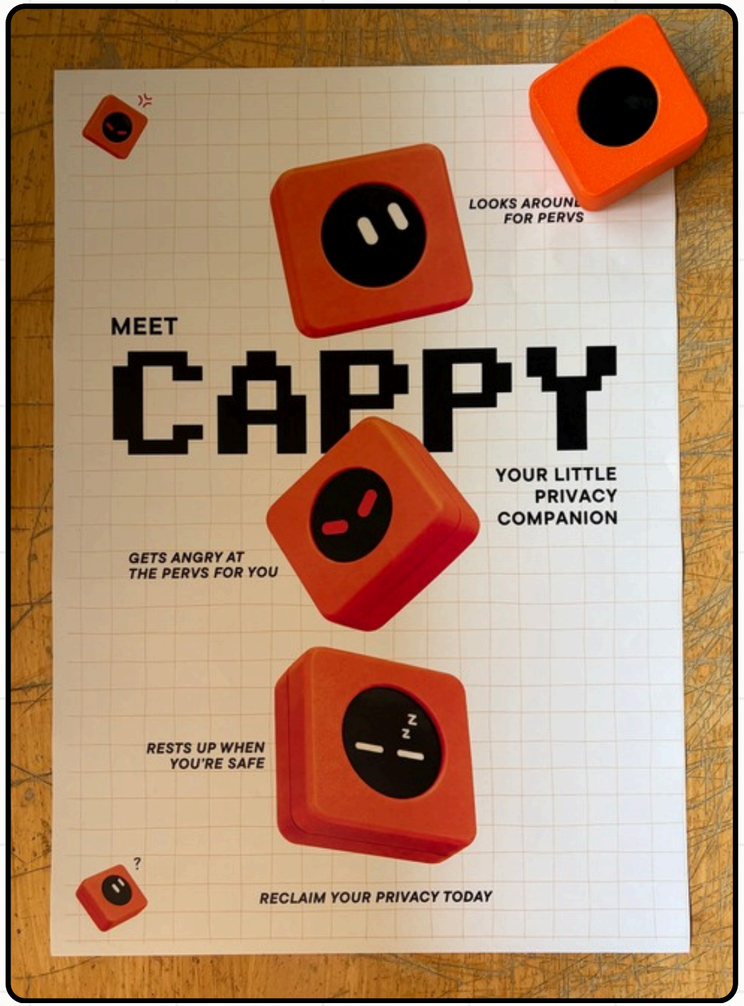
unreleased upcoming meta rayban glasses

[Link to a closer look at the glasses]

Final Poster

After I had my product ready and my glasses procured, I had one last artefact left for my presentation- the poster. I wanted it to be simple- minimal on text, while still encapsulating the functionality and personality of Cappy, making sure that he's the limelight of the show.





Final Presentation

Finally, the day has come, it was time for Cappy to shine and speak for itself. I structured my presentation in a way that it would get to speak for itself. A Minimal “booth” with my poster and Cappy on my body was all I needed.

The demo

I demoed CAP in front of the class by calling in a seasoned pervert (Sudarsinii) wearing my special version of the meta glasses as Cappy’s expressions changed and the vibration motor activated in real time as she got closer to me. Once the pervert was gone and cappy started to doze off I demoed the knock detection by knocking at the device to wake cappy up.

Watch the demo here (pervert detection)



test run before presentation



presenting the project

Feedback and observations

While presenting my project, I noticed some issues with the device that could be worked upon-

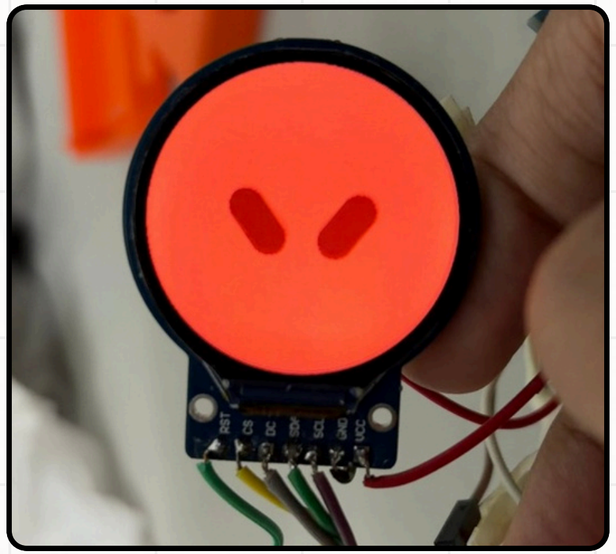
- Given the size and weight of Cappy, it would always sag a little, meaning I had to frequently adjust it or hold it up to make the screen face towards me instead of downwards. I need to do more work on the locking mechanism for Cappy to make it more robust
- The range at which Cappy goes from curious to angry can vary a lot (3.5m on the high end and <1m on the low end) depending on the room you're in and the amount of interfering Bluetooth devices around you. I hope to tweak the code to make this range more consistent going forward.
- My presentation in its current state is just a concept. The concept needs to be tested with the real Meta glasses for them to actually be considered more than a promise.



Fun Last minute add ons

Rage mode

I coded in and added rage mode to cappy as a last-minute decision. rage mode only activated when someone with the glasses tries to come even closer to you after Cappy is angry at them



New ways to wear the device

One of the philosophies of Cappy is user freedom. The user should be allowed to use Cappy and confront glassholes however they want and that freedom should also extend to how they wear it. if somone wants everyone around them to meet and interact with cappy they can wear him on their shirt as usual but if someone wants to wear it in a more discrete manner, cappy also functions well as a companion living on the waistband of your pants.





Reflections and the future

Working on CAP and Cappy has been one of the most intensive and transdisciplinary projects I have worked on at Srishti up till now and there are a lot of learning outcomes and reflections from it:

- While I was already aware of how art and politics go hand in hand, the practice of critical making takes the connections even further with the conversations it sparks. Never before have I been so enthusiastic to share my project with everyone around me and never before has that enthusiasm been reciprocated with conversations around disdain for the protocols I am trying to fight with the help of Cappy.
- While I enjoy the conversations Cappy can spark in its current state, I want to take it a step further and iterate on Cappy more to make it a feasible product for the people around me. I do not feel like I'm done with this project, I still have so many Ideas waiting to be executed that I can work on either personally or take on for my capstone next semester.
- **Do not** use superglue as an alternative for solder in emergency situations (long story)

THANK YOU